

Sun-powered laptops

In July this year, WeWi Telecommunications Inc., an Ontario-based company, unveiled a solar-powered laptop designed to bypass the electricity grid. The product will be especially useful in rural areas of developing countries, where power supply is unreliable.

WeWi's founders came up with the idea in early 2012, after visiting Ghana on business. Over the next 14 months, they designed Sol—a solar-powered laptop. The first order of around 20,000 units, priced around \$300, is likely to ship within the next few months. There have been solar-powered laptops in the past, including one from Samsung that was launched in 2010. However, these devices did not take off as they took too long to charge. WeWi hopes to have overcome these problems with its efficient design. It has priced the laptop economically hoping to promote the segment. If successful, the company hopes to build a subsequent version with satellite connectivity—another feature that will be very useful in rural areas that do not have proper landline infrastructure.

What's inside: Sol runs on a four-panel solar-charging device built into the computer's case. It takes approximately two hours to charge the laptop's battery, which then lasts for about eight to ten hours. The laptop can run off the solar panels directly when exposed to sunlight, bypassing the battery. Although the computer is slightly heavier than today's laptops, it is still not too tough to carry and fits into most laptop bags. The device is sturdy and robust, with rubber lining on the sides and a thick outer shell. The solar panels are housed inside a case on the back of the screen lid, and are not noticeable when folded up. The devices will come with a 250GB hard drive and 4GB memory, as well as a suite of free office and accounting software.

Web: <http://mobile.solalaptop.com>



Device that detects bladder cancer

Several studies published in the last decade have suggested that chemicals/metabolite levels visible in our breath can indicate the risks for diseases such as stomach cancer and heart failure, and that the scent of gases in urine could be used to test for bladder cancer. Building on such findings and their earlier attempts at building an electronic nose, a team from the University of Liverpool and the University of the West of England (Bristol) recently disclosed a new invention, the Odoreader. The device uses sensors to sniff out the chemicals in urine and delivers an accurate bladder cancer diagnosis within 30 minutes.

What's inside: The Odoreader is a volatile organic compound (VOC) sensor, which detects chemicals in gases present. The team trained it to look for the relevant chemicals using 98 different urine samples. It was then tested using samples from 24 patients suffering from bladder cancer, and 74 samples from those with other non-cancerous bladder problems. Using just nine biomarkers, the system correctly diagnosed all the cancer patients and 70 out of 74 cases where cancer was not present. The results are very promising, but the 6 per cent error rate is yet to be overcome.

In a paper on Odoreader published in PLOS One (www.plosone.org), the authors mention that “the small sample size does not allow a proper statistical assessment of the effects of other confounding factors such as smoking status, drug treatments, ethnicity.” Hence the researchers plan to work on larger samples to test the device further before it can be used in hospitals.

Web: <http://news.liv.ac.uk/2013/07/09/scent-device-could-help-detect-bladder-cancer>

